

Math 61: Introduction to Discrete Structures

Discussion 1: Sets and Functions

1 Introduction to Sets

1. What is the cardinality of the following sets?

- (i) $\emptyset$
- (ii) $\{\emptyset\}$
- (iii) $\{a, b, a, c\}$
- (iv) $\{\{a\}, \{a, b\}, \{a, a\}, a, b\}$

2. Draw the Venn diagram for the following sets.

- (i) $A \cap \overline{B}$
- (ii) $B \cup (B - A)$
- (iii) $(A \cup B) - B$
- (iv) $B \cap \overline{(C \cup A)}$

3. If X is a set such that $|X| = 10$, how many elements does $\mathcal{P}(X)$ have? How many proper subsets does X have? What about for $|X| = n$?

4. Let $X = \{n \in \mathbb{Z} | n \geq 2\}$. For $k \geq 2$, let $X_k = \{kn : n \in X\}$. What is the following set?

$$X - \bigcup_{k=2}^{\infty} X_k$$

5. Let A, B be two sets. Prove that the intersection $A \cap B \subset A$, and that $B \subset A \cup B$.

6. Let A be a set. Prove that $\overline{\overline{A}} = A$. ($\overline{A}$ is the complement of A)

2 Introduction to Functions

1. Which of the following are functions $f : \{1, 2, 3, 4\} \rightarrow \{a, b, c, d\}$? If so, is it injective? What about surjective?
 - (i) $\{(1, a), (2, a), (3, c), (4, b)\}$
 - (ii) $\{(1, c), (2, a), (3, b), (4, c), (2, d)\}$
 - (iii) $\{(1, c), (2, d), (3, a), (4, b)\}$
 - (iv) $\{(1, d), (2, d), (4, a)\}$
 - (v) $\{(1, b), (2, b), (3, b), (4, b)\}$
2. Given $g = \{(1, b), (2, c), (3, a)\}$, a function from $X = \{1, 2, 3\}$ to $Y = \{a, b, c, d\}$, and $f = \{(a, x), (b, x), (c, z), (d, w)\}$, a function from Y to $Z = \{w, x, y, z\}$, write $f \circ g$ as a set of ordered pairs. Also, draw the arrow diagram for $f \circ g$.
3. For a function $f : X \rightarrow Y$, we say that a function $g : Y \rightarrow X$ is a *left inverse* to f if $g \circ f$ is the identity on X , i.e., for all $x \in X$, $g \circ f(x) = x$. If f has a left inverse, is f necessarily injective? Is f necessarily surjective?
4. For a function $f : X \rightarrow Y$, we say that a function $g : Y \rightarrow X$ is a *right inverse* to f if $f \circ g$ is the identity on Y , i.e., for all $y \in Y$, $f \circ g(y) = y$. If f has a right inverse, is f necessarily injective? Is f necessarily surjective?
5. Let A, B, C be sets and $f : A \rightarrow B$, $g : B \rightarrow C$ be functions. Define $h = g \circ f$.
 - (a) Show that if h is injective, then f is injective. Show that g may not necessarily be injective
 - (b) Show that if h is surjective, then g is surjective. Show that f may not necessarily be surjective.